

AWS Cloud Curriculum — Services & Tools

1 DevOps & AWS Learning Hub

Complete handbooks with **all chapters written** and downloadable PDFs.

The page numbers in each handbook **index match that PDF's footer** (not estimates).

Resource	Chapters	This edition	PDF
DevOps Handbook	60	298 pages	devops-handbook.pdf
AWS Handbook	41	214 pages	aws-handbook.pdf
DevOps Curriculum	reference	9 pages	devops-curriculum.pdf
AWS Curriculum	reference	9 pages	aws-curriculum.pdf
Combined guide	all of the above	531 pages	devops-aws-complete-guide.pdf

1.1 Download PDFs

Pre-built PDFs are in the [pdf/](#) folder. On GitHub: browse to the file and click **Download**.

PDF	Description
devops-handbook.pdf	60 chapters, 298 pages (index matches footer)
aws-handbook.pdf	41 chapters, 214 pages (index matches footer)
devops-aws-complete-guide.pdf	All content, 531 pages — see combined index
devops-curriculum.pdf	DevOps tools & concepts (9 pages)
aws-curriculum.pdf	AWS services list (9 pages)

Rebuild PDFs after editing markdown:

```
./scripts/build-pdfs.sh
```

Requires `pandoc` and `wkhtmltopdf`.

1.2 Handbook coverage

1.2.1 DevOps Handbook (60 chapters, 12 parts)

1. Introduction & Mindset
2. Linux & Shell
3. Git & Collaboration
4. Networking
5. Scripting & Programming

6. Infrastructure as Code (Terraform, Ansible, Packer)
7. Containers (Docker, Compose, registries)
8. Kubernetes (architecture, workloads, Helm, GitOps)
9. CI/CD (GitHub Actions, Jenkins, GitLab, DORA)
10. Observability & SRE (Prometheus, ELK, OpenTelemetry, chaos)
11. Security / DevSecOps (SAST, Vault, SBOM, Falco)
12. Advanced (platform engineering, FinOps, MLOps, capstone)

1.2.2 AWS Handbook (41 chapters, 10 parts)

1. Cloud Foundations
2. IAM & Organizations
3. Networking (VPC, ALB, Route 53, TGW)
4. Compute (EC2, Lambda, ECS, EKS)
5. Storage (S3, EBS, EFS, Backup)
6. Databases (RDS, DynamoDB, ElastiCache)
7. Application Integration (API Gateway, SQS, Step Functions)
8. DevOps on AWS (CloudFormation, CDK, CodePipeline)
9. Observability (CloudWatch, X-Ray, CloudTrail)
10. Security & Well-Architected capstone

1.3 Suggested study order

Linux + Git → Networking → Python/Bash → Terraform → Docker → Kubernetes
→ CI/CD → Observability → Security → AWS (parallel or after cloud basics)

Each chapter includes concepts, commands, examples, hands-on labs, and review questions.

2 AWS Cloud Curriculum: Services & Tools (Beginner → Advanced)

All major AWS services grouped by domain, with learning priority.

Legend: ★ = core (learn first) · ◆ = intermediate · ◇ = advanced/specialized

2.1 1. Account, Identity & Governance ★

Service / Tool	Purpose
AWS Organizations ★	Multi-account structure, SCPs
IAM ★	Users, groups, roles, policies, MFA
IAM Identity Center (SSO) ★	Workforce federation, permission sets
AWS Control Tower ◆	Landing zone automation
AWS Config ◆	Resource compliance & history
AWS CloudTrail ★	API audit logging
AWS Service Catalog ◇	Governed self-service provisioning
AWS Resource Access Manager (RAM) ◆	Cross-account resource sharing
AWS Account Factory ◆	Automated account vending (Control Tower)
AWS CLI / CloudShell ★	Command-line & browser shell
AWS Management Console ★	Web UI
AWS CloudFormation ★	Infrastructure as Code (native)
AWS CDK ◆	IaC in TypeScript/Python/etc.
Terraform + AWS Provider ★	Industry-standard IaC
AWS Tagging & Resource Groups ★	Cost & ops organization

2.1.1 Concepts

- Shared responsibility model
 - Root account hygiene (MFA, no daily use)
 - Least privilege & permission boundaries
 - SCP vs IAM policy
 - Cross-account roles & assume-role
 - AWS global vs regional services
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2.2 2. Networking & Content Delivery ★

Service	Purpose
VPC ★	Virtual network, subnets, route tables
Internet Gateway / NAT Gateway ★	Public & private egress
VPC Endpoints (Gateway & Interface) ◆	Private AWS API access

Security Groups & NACLs ★	Stateful vs stateless firewalls
Elastic IP ★	Static public IPs
Route 53 ★	DNS, health checks, routing policies
CloudFront ★	CDN, edge caching
ALB / NLB / GWLB ★	Layer 7, 4, and gateway load balancers
VPC Peering / Transit Gateway ◆	Inter-VPC & hybrid connectivity
AWS VPN / Direct Connect ◆	On-prem hybrid networking
PrivateLink ◆	Private service exposure
AWS Global Accelerator ◇	Anycast for global apps
Route 53 Resolver ◆	Hybrid DNS
AWS Network Firewall ◇	Managed perimeter firewall
VPC Lattice ◇	Service-to-service networking

2.2.1 Tools

- **Reachability Analyzer, VPC Flow Logs, Traffic Mirroring**

2.3 3. Compute ★

Service	Purpose
EC2 ★	Virtual machines, AMIs, instance types
Auto Scaling Groups ★	Horizontal scaling
Elastic Beanstalk ◆	PaaS for web apps
Lambda ★	Serverless functions
ECS / Fargate ★	Container orchestration (AWS-native)
EKS ★	Managed Kubernetes
AWS Batch ◆	Batch/HPC jobs
Lightsail ◆	Simplified VPS
App Runner ◆	Container PaaS
EC2 Image Builder ◆	Automated AMI pipelines
AWS Outposts / Wavelength / Local Zones ◇	Edge & hybrid compute
Serverless Application Repository ◇	Share Lambda apps

2.3.1 Tools

- **AWS Systems Manager** (Session Manager, Run Command, Patch Manager)
- **EC2 Instance Connect**

2.4 4. Storage ★

Service	Purpose
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S3 ★	Object storage, versioning, lifecycle
EBS ★	Block storage for EC2
EFS ♦	Managed NFS
FSx ♦	Windows/Lustre/ONTAP/OpenZFS file systems
S3 Glacier ♦	Archive tiers
Storage Gateway ◇	Hybrid storage
AWS Backup ♦	Centralized backup
DataSync / Transfer Family ♦	Large data migration

2.4.1 Concepts

- Storage classes (S3 Standard, IA, Glacier)
- Encryption at rest (SSE-S3, SSE-KMS, SSE-C)
- S3 bucket policies vs ACLs
- EBS volume types (gp3, io2, st1)

2.5 5. Databases ★

Service	Purpose
RDS ★	Managed MySQL, PostgreSQL, MariaDB, Oracle, SQL Server
Aurora ★	AWS-native relational, serverless option
DynamoDB ★	Serverless NoSQL key-value/document
ElastiCache ♦	Redis / Memcached
DocumentDB ♦	MongoDB-compatible
Neptune ◇	Graph database
Timestream ◇	Time-series
Keyspaces ◇	Cassandra-compatible
Redshift ♦	Data warehouse
QLDB ◇	Ledger database
MemoryDB for Redis ♦	Durable Redis
RDS Proxy ♦	Connection pooling

2.6 6. Application Integration & Messaging ★

Service	Purpose
SQS ★	Managed message queues
SNS ★	Pub/sub notifications
EventBridge ★	Event bus, scheduling
Step Functions ♦	Workflow orchestration

MQ ♦	Managed ActiveMQ/RabbitMQ
AppSync ♦	GraphQL API
API Gateway ★	REST/WebSocket/HTTP APIs

2.7 7. Developer Tools & CI/CD ★

Service	Purpose
CodeCommit ♦	Git repos (legacy; many use GitHub)
CodeBuild ★	Managed build
CodeDeploy ♦	Deployment automation
CodePipeline ★	CI/CD orchestration
CodeArtifact ♦	Package repository
Cloud9 ♦	Cloud IDE
CDK / SAM / Copilot ♦	IaC & serverless/container tooling
X-Ray ♦	Distributed tracing
CloudWatch Synthetics ♦	Canary monitoring

2.8 8. Observability & Management ★

Service	Purpose
CloudWatch ★	Metrics, alarms, dashboards, logs
CloudWatch Logs Insights ★	Log query
AWS X-Ray ♦	Tracing
Systems Manager ★	Ops hub (Parameter Store, Patch, Session)
CloudTrail ★	Audit
AWS Health Dashboard ★	Service events
Trusted Advisor ♦	Best-practice checks
Personal Health Dashboard ★	Account-specific events
Service Quotas ♦	Limit management
OpsWorks ◇	Legacy Chef/Puppet (avoid for new projects)

2.9 9. Security, Identity & Compliance ★

Service	Purpose
IAM ★	(see section 1)
KMS ★	Encryption key management
Secrets Manager ★	Rotating secrets

Certificate Manager (ACM) ★	TLS certificates
WAF & Shield ♦	Web firewall & DDoS
GuardDuty ★	Threat detection
Inspector ♦	Vulnerability scanning
Macie ♦	S3 data classification
Security Hub ♦	Central security findings
Detective ◇	Security investigation
Firewall Manager ♦	Central WAF/rule management
Artifact ♦	Compliance reports
Audit Manager ◇	Continuous audit evidence
CloudHSM ◇	Dedicated HSM
Cognito ♦	User pools & identity for apps
Verified Permissions ◇	Cedar policy engine
IAM Access Analyzer ♦	External access findings

2.10 10. Migration & Transfer ♦

Service	Purpose
Migration Hub	Migration tracking
Application Migration Service (MGN)	Lift-and-shift
Database Migration Service (DMS)	DB replication
Server Migration Service	(legacy → MGN)
Transfer Family	SFTP/FTPS/FTP to S3
Snow Family	Physical data transfer appliances

2.11 11. Analytics & Big Data ♦

Service	Purpose
Athena ♦	SQL on S3
Glue ♦	ETL & data catalog
EMR ♦	Hadoop/Spark
Kinesis ♦	Streaming (Data Streams, Firehose, Analytics)
QuickSight ♦	BI dashboards
Lake Formation ♦	Data lake governance
MSK ♦	Managed Kafka
OpenSearch Service ♦	Search & analytics (Elasticsearch fork)
Data Pipeline ◇	Legacy orchestration

Clean Rooms ◇	Privacy-preserving analytics
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2.12 12. Machine Learning & AI ◇

Service	Purpose
SageMaker ◇	End-to-end ML platform
Bedrock ◇	Foundation models API
Comprehend / Rekognition / Textract / Polly / Transcribe ◇	AI APIs
Forecast / Personalize / Kendra ◇	Specialized ML
CodeWhisperer / Q Developer ◆	AI-assisted development

2.13 13. IoT, Media & Specialized ◇

Service	Purpose
IoT Core / Greengrass	Device connectivity
MediaConvert / Live / Package	Video processing
GameLift	Game server hosting
WorkSpaces / AppStream	Virtual desktops
Connect	Contact center

2.14 14. Cost & Billing ★

Service / Tool	Purpose
Cost Explorer ★	Spend analysis
Budgets ★	Alerts & limits
Cost & Usage Report (CUR) ◆	Detailed billing export
Pricing Calculator ★	Estimate costs
Savings Plans / Reserved Instances ◆	Commitment discounts
Compute Optimizer ◆	Right-sizing recommendations
Billing Conductor ◇	Reseller/MSP billing

2.14.1 Concepts

- Pay-as-you-go vs reserved vs spot
- Tagging strategy for cost allocation
- FinOps culture & unit economics

2.15 15. Well-Architected Framework (conceptual ★)

Six pillars — learn alongside every service:

1. **Operational Excellence** — run & monitor systems
2. **Security** — protect data & assets
3. **Reliability** — recover from failure
4. **Performance Efficiency** — use resources well
5. **Cost Optimization** — avoid unnecessary spend
6. **Sustainability** — minimize environmental impact

Tool: AWS Well-Architected Tool (workload reviews)

2.16 Certification path (recommended order)

Order	Certification	Focus
1	Cloud Practitioner (CLF)	Broad AWS vocabulary
2	Solutions Architect Associate (SAA)	Design & core services
3	Developer Associate (DVA)	CI/CD, serverless, APIs
4	SysOps Administrator (SOA)	Operations & monitoring
5	DevOps Engineer Professional (DOP)	Advanced CI/CD & automation
6	Solutions Architect Professional (SAP)	Complex multi-account design
7	Security Specialty	Deep security services
8	Networking Specialty	VPC, hybrid, TGW

2.17 Hands-on lab progression

1. Create IAM user, MFA, CLI profile
2. VPC with public/private subnets + NAT
3. EC2 + ALB + Auto Scaling
4. S3 static site + CloudFront + Route 53
5. RDS Multi-AZ + Secrets Manager
6. Lambda + API Gateway + DynamoDB
7. ECS Fargate or EKS cluster with CI/CD
8. CloudWatch dashboards + alarms + X-Ray
9. Terraform modules for reusable infra
10. Multi-account landing zone (Organizations + Control Tower)